

Siming: Five-Layer Behavioral Grammar Detection Architecture

eBPF Events
(tracee / falco)

TinyGPT Prior

0.88M params
7-VM universal

8-Token

Discretization

P0: Adaptive High-Risk

Rebirth / Disguise C2 / SUID privesc

P1: Pattern Match

ATT&CK patterns + Prototypes (96.2%)

P2: Context Anomaly

Per-slot NLL > τ (PARENT/DST/DT)

P3: Rarity

PROC/ARGV rarity + <UNK> detection

P4: Temporal (2nd order)

CV < 1.5 / tail_ratio / eps_corr

P5: Adaptive Low-Risk

Sleep stepping / Recon uniform

Alert + ATT&CK Technique + Severity + Automated Response

Self-Learning

Embed → Cluster

(K=22, sil=0.653)

→ Pattern Extract

→ Human Review

→ patterns.jsonl